

JINING, China

WMO: 534800

Lat: 41.0295N

Lon: 113.0704E

Elev: 1416

StdP: 85.43

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.7	-20.7	-31.2	0.2	-17.1	-29.1	0.3	-16.1	10.5	-12.9	9.2	-12.2	1.3	290	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	29.4	15.9	27.9	15.4	26.5	15.0	19.3	24.3	18.4	23.5	17.5	22.8	3.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	15.2	21.4	16.7	14.2	20.8	15.6	13.2	20.3	61.7	24.2	58.3	23.6	55.1	22.6	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	7.9	6.7	DB	-25.8	31.6	1.8	1.6	-27.1	32.7	-28.1	33.6	-29.1	34.5	-30.4	35.7
			WB	-26.2	20.7	1.7	1.3	-27.4	21.6	-28.4	22.4	-29.4	23.1	-30.7	24.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-12.0	-7.9	-0.9	7.2	13.3	18.1	20.5	18.7	13.4	6.1	-2.6	-9.9	(6)	
(7)		DBStd	12.00	4.36	4.99	5.21	5.16	4.12	3.00	2.39	2.55	3.62	4.32	4.99	4.33	(7)	
(8)		HDD10.0	2796	682	502	339	113	16	1	0	0	12	134	379	618	(8)	
(9)		HDD18.3	4878	940	735	597	335	162	39	7	26	152	381	629	877	(9)	
(10)		CDD10.0	1113	0	0	1	28	119	243	326	271	113	12	0	0	(10)	
(11)		CDD18.3	154	0	0	0	1	6	32	74	38	3	0	0	0	(11)	
(12)		CDH23.3	1401	0	0	0	12	102	331	622	294	40	0	0	0	(12)	
(13)		CDH26.7	279	0	0	0	1	14	66	150	43	6	0	0	0	(13)	
(14)	Wind	WSAvg	2.4	2.2	2.5	2.9	3.2	3.0	2.4	1.9	1.7	2.0	2.4	2.5	2.5	(14)	
(15)	Precipitation	PrecAvg	361	2	3	7	17	24	53	90	102	40	16	5	2	(15)	
(16)		PrecMax	535	10	20	25	80	59	138	178	202	103	40	16	7	(16)	
(17)		PrecMin	168	0	0	0	0	3	5	26	19	3	1	0	0	(17)	
(18)		PrecStd	93	2	4	6	16	13	29	41	53	22	12	4	2	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	7.8	16.7	24.9	28.6	30.6	31.6	29.9	27.1	21.2	12.6	3.2	(19)	
(20)		MCWB	-2.7	0.8	5.5	9.9	12.8	15.0	17.3	16.4	13.8	10.5	4.3	-2.0		(20)	
(21)		2%	DB	-0.7	4.8	13.1	21.7	26.1	28.5	29.8	27.8	24.2	18.3	9.3	0.5	(21)	
(22)		MCWB	-4.8	-1.1	3.7	8.4	11.4	14.3	16.9	16.4	12.9	8.8	2.6	-3.6		(22)	
(23)		5%	DB	-2.9	2.4	10.3	19.0	24.0	27.0	28.3	26.3	22.3	16.0	7.0	-1.5	(23)	
(24)		MCWB	-6.3	-2.6	2.1	7.1	10.3	13.6	16.4	15.9	12.3	7.3	1.2	-4.9		(24)	
(25)		10%	DB	-4.8	0.4	7.9	16.4	22.0	25.2	26.8	24.9	20.6	13.9	4.7	-3.2	(25)	
(26)		MCWB	-7.6	-3.6	0.8	5.9	9.6	13.3	16.1	15.5	11.5	6.5	0.1	-6.1		(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.4	1.3	6.2	11.6	14.4	18.4	21.1	20.3	17.4	11.7	5.4	-1.4	(27)	
(28)		MCDWB	1.8	6.6	14.4	21.0	23.7	23.9	26.1	25.1	22.0	18.5	10.0	2.5		(28)	
(29)		2%	WB	-4.6	-0.6	4.1	9.4	12.9	17.0	19.7	19.1	15.2	10.1	3.2	-3.2	(29)	
(30)		MCDWB	-1.0	3.7	11.7	18.5	22.3	23.3	24.9	23.8	20.3	16.4	8.6	-0.1		(30)	
(31)		5%	WB	-6.1	-2.1	2.6	8.0	11.8	16.0	18.8	18.2	13.8	8.7	1.8	-4.6	(31)	
(32)		MCDWB	-3.0	2.1	9.7	16.9	20.5	22.7	24.1	23.0	19.3	14.0	6.1	-1.7		(32)	
(33)		10%	WB	-7.5	-3.5	1.1	6.7	10.7	15.1	17.9	17.2	12.7	7.4	0.4	-6.1	(33)	
(34)		MCDWB	-4.8	0.4	7.3	14.8	19.3	21.8	23.2	22.2	18.5	12.5	4.5	-3.2		(34)	
(35)	Mean Daily Temperature Range	MDBR	10.8	11.4	11.8	12.2	12.2	11.3	10.3	10.6	11.3	11.1	10.2	9.8		(35)	
(36)		5% DB	MCDDBR	12.7	13.6	14.7	15.7	15.5	14.3	13.2	12.8	13.8	14.2	12.8	11.7		(36)
(37)		MCWBR	9.7	9.3	8.7	7.9	6.8	5.4	4.4	4.8	5.9	7.1	7.8	8.7		(37)	
(38)		5% WB	MCDDBR	12.3	12.5	14.0	13.8	12.8	10.8	9.7	9.5	11.1	12.1	11.8	11.4		(38)
(39)		MCWBR	9.5	8.9	8.8	7.5	6.4	4.7	4.5	4.4	5.7	6.9	7.5	8.7		(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.287	0.344	0.380	0.437	0.480	0.535	0.505	0.496	0.449	0.387	0.319	0.295		(40)	
(41)		taud	2.266	2.109	2.049	1.965	1.899	1.829	1.963	1.959	2.008	2.112	2.239	2.260		(41)	
(42)		Ebn at Noon	861	847	858	833	805	760	779	772	778	786	808	815		(42)	
(43)		Edh at Noon	91	126	150	176	193	207	180	175	156	124	94	85		(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.54	3.49	4.89	5.91	6.57	6.09	5.92	5.55	4.70	3.78	2.68	2.20		(44)	
(45)		RadStd	0.23	0.30	0.40	0.37	0.42	0.49	0.29	0.35	0.26	0.21	0.29	0.18		(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-2.35	N/A	-0.54	-0.82	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.42	N/A	N/A	N/A	N/A

Nomenclature: See separate page